

Wildfire Resiliency Code



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COUNCIL BILL NO. 26-
ORDINANCE NO.

AN ORDINANCE ENACTING A NEW CHAPTER 106 OF THE CITY CODE OF THE CITY
OF ARVADA, COLORADO RELATING TO THE ESTABLISHMENT OF THE ARVADA
WILDFIRE RESILIENCY CODE

BE IT ORDAINED BY THE CITY COUNCIL OF THE CITY OF ARVADA, COLORADO:

Section 1. Chapter 106, Arvada Wildfire Resiliency Code, is hereby enacted and shall read as follows:

Chapter 106

“ARTICLE I. IN GENERAL

Sec. 106-1. Rules of construction.

- (a) *Scope.* Unless otherwise expressly stated, the following words and terms shall, for the purposes of this Code, have the meanings indicated in this chapter.
- (b) *Interchangeability.* Words stated in the present tense include the future; words stated in the masculine gender include the feminine and neuter; And the singular number includes the plural and the plural the singular.
- (c) *Terms defined in other codes.* Where terms are not defined in this code and are defined in other codes and regulations, such terms shall have the meaning ascribed to them as in those codes and regulations.
- (d) *Terms not defined.* Where terms are not defined through the methods authorized by this section, such terms shall have ordinarily accepted meanings such as the context implies.
- (e) *Defined terms in this code.* Where terms defined by this code are used within this code, the term is italicized.

Sec. 106-2. Definitions.

The following words, terms, and phrases, when used in this chapter, shall have the meanings ascribed to them in this section, except where the context clearly indicates a different meaning:

Accessory Structure means a subordinate building, or portion of a main building, the use of which is incidental to that of the main building on the same lot. This includes detached garages, workshop, sheds, barns, coops, greenhouses, and detached living space.

Approved means acceptable to the *code official*.

Barn means a building used exclusively for the housing and care of horses or other permitted livestock, and for the storage of feed, hay, other farm crops, and farm or equine equipment. Barns may not be used for a garage or as a storage shed for items not associated with livestock or farm crops.

Building means any *structure* intended for supporting or sheltering any occupancy.

Class A Tests means tests applicable to *roof coverings* that are expected to be effective against severe fire exposure, before a high degree of fire protection to the *roof deck*, do not slip from position, and are not expected to present a flying brand hazard.

Code Official means the official designated by the jurisdiction to interpret and enforce this code, or the *code official's* authorized representative.

Defensible Space means an area either natural or man-made, where material capable of allowing a fire to spread unchecked has been treated, cleared or modified to slow the rate and intensity of an advancing wildfire and to create an area for fire suppression operations to occur.

Embellishments means elements incorporated in design and construction for ornamental or decorative purposes that are not integral to the *structure* or structural support.

Fire Intensity Classification means the level of fire intensity identified for areas where significant fuel hazards and associated dangerous fire behavior may exist, based upon vegetative fuels, topography, weather conditions, and flame length value.

Fire-Resistance-Rated Construction means the use of materials and systems in the design and construction of a *building* or *structure* to safeguard against the spread of fire within a *building* or *structure* and the spread of fire to or from *buildings* or *structures* to the *wildland-urban interface* area.

Fire-Retardant-Treated Wood means any wood product that, when impregnated with chemicals by a pressure process or other means during manufacture, shall have, when tested in accordance with ASTM E84 or UL 723, a listed *flame spread index* of 25 or less. The ASTM E84 or UL 723 test shall be continued for an additional 20 minute. And the flame front shall not progress more than 10.5 feet beyond the center line of the burners at any time during the test.

Flame Spread Index means comparative measure, expressed as the dimensionless number, derived from visual measurements of the spread of flame versus time for a material tested in accordance with ASTM 84.

Fuel Modification means a method of modifying fuel load by reducing the amount of non fire resistive vegetation or altering the type of vegetation to reduce the fuel load.

Habitable Space means a space in a *building* for living, sleeping, eating or cooking.

Historic Structure means any *building* or *structure* that has one or more of the following:

1. Listed, or certified as eligible for listing, by the state historic preservation office or Keeper of the National Register of Historic Places, in the National Register of Historic Places.
2. Designated as historic under applicable state or local law.
3. Certified as contributing resource within the National Register, state designated or locally designated historic district.

Home Ignition Zone (Hiz) means the home and the area around the home which takes into account both the potential of the *structure* to ignite and the quality of *defensible space* surrounding it.

Ignition-Resistant Building Material means a type of *building* material that resists ignition or sustained flaming combustion sufficiently so as to reduce losses from wildfire exposure of burning embers and small flames.

Ignition-Resistant Building Materials shall comply with any of the following:

1. Material shall be tested on all sides with the extended ASTM E84 (UL 723) test or ASTM E2768, except panel products shall be permitted to test only the front and back faces. Panel products shall be tested with a ripped or cut longitudinal gap of 1/8 inch. Materials that, when tested in accordance with the test procedures set forth in ASTM E84 or UL 723 for a test period of 30 minutes, or with ASTM E2768, comply with the following:
 - a. Flame spread. Material shall exhibit a *flame spread index* not exceeding 25 and shall not show evidence of progressive combustion following the extended 30-minute test.
 - b. Flame front. Material shall exhibit a flame front that does not progress more than 10.5 feet beyond the centerline of the burner at any time during the extended 30-minute test.
 - c. Weathering. *Ignition-resistant building materials* shall maintain their performance in accordance with this section under conditions of use. Materials shall meet the performance requirements for weathering (including exposure to temperature, moisture and ultraviolet radiation) contained in the following standards, as applicable to the materials and the conditions or use:
 - i. Method A “Test method for Accelerated Weathering of *Fire-Retardant-Treated* Wood for Fire Testing” in ASTM D2898, for *fire-retardant treated wood*, wood-plastic composite and plastic lumber materials.
 - ii. ASTM D7032 for wood-plastic composite materials.
 - iii. ASTM D6662 for plastic lumber materials.
 - d. Identification. Materials shall bear identification showing the fire test results.

Exception: Materials composed of a combustible core and a *noncombustible* exterior covering made from either aluminum at a minimum 0.019 inch (0.48 mm) thickness or corrosion-resistant steel at a minimum 0.0149 inch (0.38 mm) thickness shall not be required to be tested with a ripped or cut longitudinal gap.

Ignition-Resistant Vegetation means plants that are less likely to readily ignite from a flame or other ignition source and produce fewer embers. While they can still be damaged by fire, their foliage and stems don't significantly contribute to the intensity of the fire.

Log Wall Construction means a type of construction in which exterior walls are constructed of solid wood members and where the smallest horizontal dimension of each solid wood member is not less than 6 inches. *Log wall construction* shall follow requirements of ICC 400.

Mass Timber Construction means as described in section 602.4 of the 2024 International Building Code. Also known as Type IV construction.

Multilayered Glazed Panels means window or door assemblies that consists of two or more independently glazed panels installed parallel to each other, having a sealed air gap in between, within a frame designed to fill completely the window or door opening in which the assembly is intended to be installed.

Noncombustible, as applied to *building* construction material, means and material that, in the form in which it is used, is one of the following:

1. Material of which no part will ignite and burn when subjected to fire.
2. Any material confirming to ASTM E136 shall be considered *noncombustible* within the meaning of this section.
3. For the purposes of this code, fire-rated gypsum board tested in accordance with ASTM C1396 with no less than a 1-hour fire-resistance-rating with fire exposure from the outside only is considered a *noncombustible* material.

Occupiable Space means a room or enclosed space designed for human occupancy in which individuals congregate for amusement, education or similar purposes or in which occupants are engaged at labor.

Roof Assembly means a system designed to provide weather protection and resistance to design loads. The system consists of a *roof covering* and *roof deck* or a single component serving as both the *roof covering* and *roof deck*. A *roof assembly* can include an underlayment, thermal barrier, ignition barrier, insulation or vapor retarder.

Roof Covering means the covering applied to the *roof deck* for weather resistance, fire classification or appearance.

Roof Deck means the flat or sloped surface not including its supporting members or vertical supports.

Slope means the variation of terrain from the horizontal; The number of feet rise or fall per 100 feet measured horizontally, expressed as a percentage.

Structure means that which is built or constructed. For the purposes of this code, fences are not considered a structure.

Structure Ignition Zone is the *structure* and the area around the *structure* (or home). The SIZ takes into account both the potential of the *structure* to ignite and the quality of *defensible space* surrounding it.

Tree Crown means the primary and secondary branches growing out from the main stem, together with twigs and foliage.

Wildfire Interface Specialist means an individual who meets the minimum training and experience requirements as required by the code official for conducting *structure ignition zone* inspections.

Wildland-Urban Interface Interior means that geographical area farther away from wildland vegetation than the *Interface Perimeter* where homes are primarily exposed to radiant heat and embers from adjacent homes as well as embers from burning vegetation. These areas have the potential for urban conflagrations.

Wildland-Urban Interface Perimeter means that geographical area adjacent to wildland vegetation where homes can be exposed to embers and radiant and convective heat from burning vegetation and adjacent homes. These areas have the potential for wildland fires to transition into urban conflagrations.

Wildland-Urban Intermix means that geographical area where the built environment intermingles with wildland vegetation and where homes are primarily exposed to embers and radiant and convective heat from burning vegetation.

Secs. 106-3 – 106-09. Reserved.”

“ARTICLE II. SCOPE AND ADMINISTRATION

Sec. 106-10. Scope and General Requirements.

- (a) *Title.* Articles 1-5 of this code shall be considered a part of the Arvada Code. These regulations shall be known as, and shall be cited as, the “Arvada Wildfire Resiliency Code” and shall be referred to herein as “this code.”
- (b) *Scope.* The provisions of this code shall apply to the construction, exterior alteration, movement, exterior repair, and replacement of any *building or structure* within the *wildland-urban interface areas* in this jurisdiction. Existing non-habitable and non-occupiable *buildings or structures*, which undergo a change of occupancy to include habitable or *occupiable space*, shall comply with the provisions of this code for new *buildings or structures*. Although not considered a structure, new fences and retaining walls are required to follow the provisions of Article 5 in this code.

Buildings, structures, or conditions in existence at the time of the adoption of this code are allowed to have their use or occupancy continued, if such condition, use or occupancy was legal at the time of the adoption of this code, provided that such continued use does not constitute a distinct danger to life or property.

When portions of existing non-habitable and non-occupiable *buildings or structures* undergo a change of occupancy which includes a residential occupancy classification or single- or two-family dwellings, per the City's adopted Building Codes and Supplements the *building or structure* shall comply with the provisions of this code for new *buildings or structures*.

Buildings or structures moved into or within the jurisdiction shall comply with the provisions of this code for new *buildings or structures*.

- (c) *Date effective.* This code shall become effective on July 1, 2026, and shall apply to *permits* applied for on or after July 1, 2026. Applications for *permits* made prior to July 1, 2026, shall be governed by the terms of the codes and regulations in effect at the time of application.
- (d) *Purpose.* The purpose of this code is to establish minimum regulations for the safeguarding of life and for property protection. Regulations in this code are intended to mitigate the risk to life and *structures* from intrusion of fire from wildland fire exposures and fire exposures from adjacent *structures* and to mitigate *structure* fires from spreading to wildland fuels. The extent of this regulation is intended to be tiered commensurate with the relative level of hazard present.

The unrestricted use of property in *wildland-urban interface areas* is a potential threat to life and property from fire and resulting erosion. Safeguards to prevent the occurrence of fires and to provide adequate fire protection facilities to control the spread of fire in *wildland-urban interface areas* shall be in accordance with this code.

The standards in this code are incorporated into and are a part of the jurisdiction's adopted *Building Codes and Supplements*, and augment the jurisdiction's fire codes, to mitigate the fire- and life-safety hazards of the *wildland-urban interface areas*.

- (e) *Retroactivity.* The provisions of the code shall apply to conditions arising after the adoption thereof, conditions not legally in existence at the adoption of this code and conditions that, in the opinion of the *code official*, constitute a distinct hazard to life or property.

Exception: Provisions of this code that specifically apply to existing conditions are retroactive.

- (f) *Additions or alterations.* Additions or alterations shall be permitted to be made to any *building or structure* without requiring the existing *building or structure* to comply with all of the requirements of this code, provided that the addition or alteration conforms to that required for a new *building or structure*.

Exception: Provisions of this code that specifically apply to existing conditions are retroactive.

Additions or alterations shall not be made to an existing *building* or *structure* that will cause the existing *building* or *structure* to be in violation of any of the provisions of this code nor shall such additions or alterations cause the existing *building* or *structure* to become unsafe. An unsafe condition shall be deemed to have been created if an addition or alteration will cause the existing *building* or *structure* to become structurally unsafe or overloaded; will not provide adequate access in compliance with the provisions of this code or will obstruct existing exits or access; will create a fire hazard; will reduce required fire resistance or will otherwise create conditions dangerous to human life.

- (g) *Maintenance.* *Buildings, structures, landscape materials, vegetation, defensible space* or other devices or safeguards required by this code shall be maintained in conformance to the code edition under which installed. The owner or the owner's authorized agent shall be responsible for the maintenance of *buildings, structures, landscape materials* and vegetation.

Sec. 106-11. Applicability.

- (a) *General.* Where there is a conflict between a general requirement and a specific requirement, the specific requirement shall govern. Where, in any specific case, different sections of this code, or any other adopted City code or regulation, specify different materials, methods of construction or other requirements, the most restrictive shall govern.
- (b) *Other Laws.* The provisions of this code shall not be deemed to nullify any provisions of local, state, or federal law.
- (c) *Application of references.* Certain additional codes and standards are referenced in this code and are listed throughout. Such codes and standards shall be considered as part of the requirements of this code to the prescribed extent of each such reference and as further regulated in the below subsections (1) and (2).
 - (1) Conflicts. Where conflicts occur between provisions of this code and the referenced codes and standards, the most restrictive shall apply.
 - (2) Provisions in referenced codes and standards. Where the extent of the reference to a referenced code or standard includes subject matter that is within the scope of this code, the provisions of this code, as applicable, shall take precedence over the provisions in the referenced standard.
- (d) *Subjects not regulated by this code.* Where applicable standards or requirements are not set forth in this code, or are contained within other laws, codes, regulations, ordinances or policies adopted by the jurisdiction, compliance with applicable standards of other nationally recognized safety standards, as *approved*, shall be deemed as prima facie evidence of compliance with the intent of this code. Nothing herein shall derogate from the authority of the *code official* to determine compliance with codes or standards for those activities or installations within the *code official's* jurisdiction or responsibility.

- (e) *Matters not provided for.* Requirements that are essential for the public safety of an existing or proposed activity, *building* or *structure*, or for the safety of the occupants thereof, which are not specifically provided for by this code, shall be determined by the *code official* consistent with the necessity to establish the minimum requirements to safeguard the public health, safety and general welfare.
- (f) *Partial Invalidity.* In the event any part or provision of this code is held to be illegal or void, this shall not have the effect of making void or illegal any of the other parts or provisions.
- (f) *Exemptions: Work exempt from the requirements of this code.* Exemptions from code requirements shall not be deemed to grant authorization for any work that violates the provisions of this code or any other laws, ordinances, regulations or rules of the jurisdiction. This code does not apply to:
 - (1) Alterations that only affect the interior of existing *structures*.
 - (2) The reconstruction, replacement, alteration, or repair of the exterior walls of an existing *building*, when less than 25 percent of the surface area of all exterior walls is affected.
 - (3) The reconstruction, replacement, alteration, or repair of the existing exterior *roof covering* of an existing *building* when less than 25 percent of the surface area of the exterior *roof covering* or an attachment thereto is affected.
 - (4) Alterations or repairs to the exterior of an existing *structure*, or an attachment to it, when less than 25 percent of the exterior of the *structure* is affected by the alteration or repair.
 - (5) Painting, staining, and similar maintenance or restorative work.
 - (6) Materials and methods of construction, when required to maintain a *building's* or *structure's* continued designation as a *historic structure*, are exempt from the requirements of this code.
 - (7) One story detached accessory, non-habitable *structures*, such as tool and storage sheds, playhouses and similar uses, provided that the floor area does not exceed 120 square feet and the structure is located greater than or equal to 10 feet from the nearest adjacent *occupiable* structure. The distance between structures is measured from the outermost point of the structures.
 - (8) *Barns* used exclusively to house livestock, store livestock feed, store farm products, protect equipment, or manage agricultural waste, located more than 50 feet from a *structure* containing *occupiable* or *habitable space*. The distance between structures is measured from the outermost point of the structures.
 - (9) Fences located more than 8 feet from the outermost point of a habitable structure.

Sec. 106-12. Code Compliance Agency.

- (a) *Creation of Agency.* The Arvada Building Safety Division and the Fire Protection Districts within Arvada are hereby designated as the code compliance agencies and the officials in charge thereof shall be known as the *code officials*. The function of the agency shall be the implementation, administration and enforcement of the provisions of this code.
- (b) *Code Official.* The *code official* is the Chief Building Official, Fire Code Official or their designee.
- (c) *Deputies.* The *code official* shall have the authority to appoint deputy *code officials*, other related technical officers, inspectors and other employees. Such employees shall have powers as delegated by the *code official*.

Sec. 106-13. Duties and Powers of the Code Official.

- (a) *Powers and duties of the code official.* The *code official* is hereby authorized to enforce the provisions of this code.
- (b) *Determination of compliance.* The *code official* shall have the authority to determine compliance with this code, to render interpretations of this code and to adopt policies and procedures in order to clarify the application of its provisions. Such interpretations, policies and procedures:
 - (1) Shall be in compliance with the intent and purpose of this code.
 - (2) Shall not have the effect of waiving requirements specifically provided for in this code.
- (c) *Technical assistance.* Documentation concerning the application of this Code and decisions or actions of the building official shall be kept in accordance with Arvada's adopted records retention schedule.
 - (1) Costs. A technical opinion and report shall be provided without charge to the jurisdiction.
 - (2) Preparer qualifications. The technical opinion and report shall be prepared by a qualified engineer, specialist, laboratory or fire safety specialty organization acceptable to the *code official*. The *code official* is authorized to require design submittals to be prepared by, and bear the stamp of, a registered design professional.
 - (3) Content. The technical opinion and report shall analyze the properties of the design, operation or use of the *building* or premises, the facilities and appurtenances situated thereon and fuel management to identify and propose necessary recommendations.
 - (4) Tests. Where there is insufficient evidence of compliance with the provisions of this code, the *code official* shall have the authority to require tests as evidence of compliance. Test methods shall be as specified in this code or by other recognized test standards. In the absence of recognized test standards, the *code official* shall approve the testing procedures. Such tests shall be performed by a party acceptable to the *code official*.

- (d) *Alternative materials, design, and methods.* The provisions of this code are not intended to prevent the installation of any material or to prohibit any design or method of construction not specifically prescribed by this code, provided that any such alternative is not specifically prohibited by this code and has been *approved*.
- (1) Approval authority. An alternative material, design or method shall be *approved* where the *code official* finds that the proposed alternative is satisfactory and complies with the below subsections (2) through (7), as applicable.
 - (2) Application and disposition. A request to use an alternative material, design or method of construction shall be submitted in writing to the *code official* for approval. Where the alternative material, design or method of construction is not *approved*, the *code official* shall respond in writing, stating the reasons the alternative was not *approved*.
 - (3) Compliance with code intent. An alternative material, design or method of construction shall comply with the intent of the provisions of this code.
 - (4) Equivalency criteria. An alternative material, design or method of construction shall, for the purpose intended, be not less than the equivalent of that prescribed in this code with respect to all of the following, as applicable:
 - (i) Quality.
 - (ii) Strength.
 - (iii) Effectiveness.
 - (iv) Durability.
 - (v) Safety, other than fire safety.
 - (vi) Fire safety.
 - (5) Tests. Tests conducted to demonstrate equivalency in support of an alternative material, design or method of construction application shall be of a scale that is sufficient to predict performance of the end use configuration. Tests shall be performed by a party acceptable to the *code official*.
 - (i) Fire tests. Tests conducted to demonstrate equivalent fire safety in support of an alternative material, design or method of construction application shall be of a scale that is sufficient to predict fire safety performance of the end use configuration. Tests shall be performed by a party acceptable to the *code official*.
 - (6) Reports. Supporting data, where necessary to assist in the approval of materials or assemblies not specifically provided for in this code, shall comply with the below subsections (i) and (ii).
 - (i) Evaluation reports. Evaluation reports shall be issued by an *approved* agency and use of the evaluation report shall require approval by the *code official* for the installation. The alternate material, design or method of construction and product evaluated shall be within the scope of the *code official's* recognition of the *approved* agency. Criteria used for the evaluation shall be identified within the report and, where required, provided to the *code official*.

- (ii) Other reports. Reports not complying with the above subsection (i) shall describe criteria, including but not limited to any referenced testing or analysis, used to determine compliance with code intent and justify code equivalence. The report shall be prepared by a qualified engineer, specialist, laboratory or fire safety specialty organization acceptable to the *code official*. The *code official* is authorized to require design submittals to be prepared by, and bear the stamp of, a registered design professional.
- (7) Peer review. The *code official* is authorized to require submittal of a peer review report in conjunction with a request to use an alternative material, design or method of construction, prepared by a peer reviewer that is *approved* by the *code official*.
- (e) *Modifications*. Where there are practical difficulties involved in carrying out the provisions of this code, the *code official* shall have the authority to grant modifications for individual cases, provided that the *code official* shall first find that one or more special individual reasons make the strict letter of this code impractical, that the modification is in conformance with the intent and purpose of this code, and that such modification does not lessen health, life and fire safety requirements. The details of the written request and action granting modifications shall be recorded and entered into the files of the code enforcement agency.
- (f) *Applications and permits*. The *code official* is authorized to receive applications, review construction documents and issue permits for construction regulated by this code, and inspect the premises for which such permits have been issued and enforce compliance with the provisions of this code.
- (g) *Right of Entry*. Where it is necessary to make an inspection to enforce the provisions of this code, or where the *code official* has reasonable cause to believe that there exists in a *structure* or on any premises a condition that is contrary to or in violation of this code that makes the *structure* or premises unsafe, dangerous or hazardous, the *code official* is authorized to enter the *structure* or premises at all reasonable times to inspect or to perform the duties imposed by this code. If such *structure* or premises is occupied, the *code official* shall present proper credentials to the occupant and request entry. If such *structure* or premises is unoccupied, the *code official* shall first make a reasonable effort to locate the owner, the owner's authorized agent or other person having charge or control of the *structure* or premises and request entry. If such entry is refused, then the *code official* shall have recourse to every remedy provided by law to secure entry.
- (1) Warrant. Where the *code official* has first obtained a proper inspection warrant or other remedy provided by law to secure entry, an owner, the owner's authorized agent, occupant or person having charge, care or control of the *structure* or premises shall not fail or neglect, after a proper request is made as herein provided, to permit entry therein by the *code official* for the purposes of inspection and examination pursuant to this code.
- (h) *Identifications*. The *code official* shall carry proper identification when inspecting *structures* or premises in the performance of duties under this code.
- (i) *Notices and orders*. The *code official* shall issue all necessary notices or orders to ensure compliance with this code. Notices of violations shall be in accordance with the violation

procedures related to the adopted Arvada Building Codes and Supplements or the Arvada Zoning Resolution, as may be applicable.

- (j) *Official records.* Official records shall be maintained in accordance with the Colorado Open Records Act and City record retention policies.
- (k) *Approved materials and equipment.* Materials, equipment and devices *approved* by the *code official* shall be constructed and installed in accordance with such approval.
 - (1) Materials and equipment reuse. Materials, equipment and devices shall not be reused unless such elements are in good working order and *approved*.

Sec. 106-14. Permits.

- (a) *General.* Within the *wildland-urban interface* Class 1 and Class 2 Areas, *structure* hardening will be regulated as part of and through the permit process specified in the adopted Arvada Building Codes and Supplements. Site and area regulations will be regulated as part of and through the *defensible space* permit process specified in the Arvada Zoning Resolution.

The code official is authorized to stipulate conditions for permits. Permits shall not be issued where public safety would be at risk, as determined by the code official.

- (b) *Permit application.* In obtaining permits, every such application shall:
 - (1) Identify the Wildland-Urban Interface Class.
 - (2) Identify and describe the work, activity, operation, practice or function to be covered by the permit for which application is made.
 - (3) Describe the land on which the proposed work is to be done by legal description, street address or similar description that will readily identify and definitely locate the proposed *building*, work, activity, operation, practice or function.
 - (4) Indicate the use or occupancy for which the proposed work is intended.
 - (5) Be accompanied by plans, diagrams, computation and specifications and other data as required to determine compliance.
 - (6) Give such other data and information as required by the *code official*.
- (c) *Preliminary inspection.* Before a permit is issued, the *code official* is authorized to inspect and approve the systems, equipment, *buildings*, devices, premises and spaces or areas to be used.
- (d) *Refusal to issue a permit.* Where the application or construction documents do not conform to the requirements of pertinent laws, the *code official* shall reject such application in writing, stating the reasons therefor.
- (e) *Validity of permit.* The issuance or granting of a permit or approval of plans, specifications and computations shall not be construed to be a permit for, or an approval of, any violation of any of the provisions of this code or of any other ordinance of the jurisdiction. Permits

presuming to give authority to violate or conceal the provisions of this code or other ordinances of the jurisdiction shall not be valid.

- (f) *Revocation of permits.* The *code official* is allowed to, in writing, suspend or revoke a permit issued under the provisions of this code whenever the permit is issued in error or on the basis of incorrect information supplied, or in violation of any ordinance or regulation or any of the provisions of this code.

Sec. 106-15. Fees.

- (a) *Payment of fees.* Fees for permits, if required, shall be in accordance with the Building Permit Fee schedule and/or the Development Application & Permit Fee schedule adopted by the city of Arvada. A permit shall not be valid until the fees prescribed by law have been paid. An amendment to a permit shall not be released until the additional fee, if any, has been paid.

Sec. 106-16. Inspections.

- (a) *General.* Inspections shall generally be conducted as specified in adopted Arvada Building Codes and Supplement, as applicable.

Sec. 106-17. Enforcement.

- (a) *General.* Enforcement of this code shall be in accordance with the adopted Arvada Building Codes and Supplements, or the Arvada Land Development code, as applicable.

Sec. 106-18. Appeals.

- (a) *General.* Appeals shall be as specified in the adopted Arvada Building Codes, the Building Code Supplements, or in the Arvada Zoning Resolution, as each may be applicable.

Generally, the Building Code Advisory Board has authority to hear and decide appeals or orders, requirements or decisions of the *code official* relating to the application and interpretation of Article 1, 2, and 4 of this code or arising out of the permitting process set forth in the adopted Arvada Building Codes and Supplements.

The Arvada Building Code Advisory Board has authority to hear and decide appeals or orders, requirements or decisions of the *code official* relating to the application and interpretation of Articles 3 and 5 of this code or arising out of the permitting process set forth in the Arvada Code.

An applicant for an appeal arising out of the administration of the provisions of this code that are incorporated into or are a part of both the Arvada Building Codes and Supplements elect to have the Building Code Advisory Board hear the entirety of the appeal.

Secs. 106-19– 106-29. Reserved.”

“ARTICLE III. WILDFIRE HAZARD IDENTIFICATION

Sec. 106-30. General.

- (a) *Scope.* Inspections shall generally be conducted as specified in adopted Arvada Building Codes and Supplement, as applicable.

Sec. 106-31. Wildland-Urban interface and designations.

- (a) *Declaration.* Inspections shall generally be conducted as specified in adopted Arvada Building Codes and Supplement, as applicable.

Sec. 106-32. Mapping and applicability.

- (a) *Applicability of code provisions.* The requirements of this code shall apply to all parcels located within designated *wildland-urban interface and intermix* areas derived from corresponding *fire intensity classifications* and ember exposure areas as identified on the Arvada Wildland-Urban Interface Overlay District map. The level of structure hardening, *defensible space*, and other mitigation measures required shall correspond to the applicable *wildland-urban interface* classification, Class 1 Intermix or Class 2 Interface.

Structures and parcels identified with low *fire intensity classification* and are therefore located in the *Interface Perimeter* and *Interface Interior* zones shall be constructed and maintained in accordance with provisions for Class 1 Interface *structure* hardening and site and area requirements.

Structures and parcels identified with moderate to high *fire intensity classifications* and therefore are in the *Intermix* zone shall be constructed and maintained in accordance with the provisions for Class 2 Intermix *structure* hardening and site and area requirements.

Sec. 106-33. Mapping amendment process.

- (a) *General.* Property owners, or their representatives, who disagree with the *wildland-urban interface* mapped areas, can submit a formal application for a map amendment pursuant to the process set forth by the *code official*.

Secs. 106-34– 106-49. Reserved.”

“ARTICLE IV. STRUCTURE HARDENING

Sec. 106-50. General.

- (a) *Scope.* Exterior design and construction of *buildings* and *structures* within the *wildland-urban interface* areas of Arvada shall be constructed in accordance with this Article.

Sec. 106-51. Class 1 interface structure hardening.

- (a) *General.* Class 1 *structure* hardening shall be in accordance with Sec. 106-51(b) through (j) and shall apply to *buildings* and *structures* hereafter constructed, modified or relocated into or within areas of the *wildland-urban interface* Class 1.
- (b) *Roofing.* Roof shall have a *roof assembly* classified as *Class A* when tested in accordance with ASTM E108 or UL790.
 - (1) Flame and ember protection of roofs. For roof assemblies where the *roof covering* profile creates a space between the *roof covering* and *roof deck*, the space shall resist the entry of flames and embers by one or more of the following methods:
 - (i) Fire stopping, with *noncombustible material*, of the space between the *roof covering* and the *roof deck*.
 - (ii) Installation of one layer of cap sheet complying with ASTM D3909 over the combustible *roof deck*.
 - (iii) Installation of a listed *Class A* classified *roof assembly*.
 - (2) Open valley flashings. Open valley flashing shall not be less than 0.019 inch (No. 26 galvanized sheet gauge) corrosion resistant metal installed over a minimum 36-inch-wide underlayment consisting of one layer of cap sheet complying with ASTM D3909 running the full length of the valley.
- (c) *Gutters and downspouts.* Gutters and downspouts shall be constructed of *noncombustible* material.
- (d) *Ventilation openings.* Ventilation openings for enclosed attics, enclosed rafter spaces, and under floor spaces shall be in accordance with Sec. 106-51(d)(1) or Sec. 106-51(d)(2) as applicable.
 - (1) Performance requirements. Ventilation openings shall be fully covered with listed vents, tested in accordance with ASTM E2886, to demonstrate compliance with all of the following requirements:
 - (i) There shall be no flaming ignition of the cotton material during the Ember Intrusion Test.
 - (ii) There shall be no flaming ignition during the Integrity Test portion of the Flame Intrusion Test.
 - (iii) The maximum temperature of the unexposed side of the vent shall not exceed 662°F (350°C).
 - (2) Prescriptive requirements. Ventilation openings for enclosed attics, enclosed rafter spaces, and under floor spaces shall be covered with *noncombustible* corrosion resistant mesh with openings not to exceed 1/8 inch.
- (e) *Exterior wall.* Exterior walls shall be limited to the following:

- (1) Vinyl siding as part of or applied to exterior wall assemblies with a 1-hour-fire-resistance rating, rated for exposure on the exterior side.
- (2) Vinyl siding over *noncombustible* sheathing.
- (3) A wall covered with *noncombustible siding*.
- (4) A wall covered with *fire-retardant-treated wood siding*.
- (5) A wall covered with *siding made of ignition resistant building materials*.
- (6) *Mass timber or log wall construction*.

Such materials shall extend from the top of the foundation to the underside of the eave or the underside of the roof sheathing. Exterior wall coverings shall be labeled for exterior use or shall be covered by a material labeled for exterior use.

Exceptions:

1. Exterior wall *embellishments* and architectural trim (exclusive of trim on exterior windows and doors) not to exceed 5 percent of the square footage of exterior walls.
 2. Roof or wall top cornice projections and similar assemblies.
 3. Solid wood rafter tails and solid wood blocking installed between rafters having a minimum dimension of 2 inches nominal.
- (f) *Flashing*. A minimum of 6 inches of metal flashing or *noncombustible* material applied vertically between the wall sheathing and the exterior cladding shall be installed at the ground, decking, and roof intersections.

Combustible sheathing products exposed by the gap created at the base of the exterior walls, posts, or columns must be protected with *noncombustible* material or *ignition-resistant building materials* while still permitting drainage and moisture control from behind exterior cladding.

- (g) *Exterior glazing*. Exterior windows, window walls and glazed doors, windows within exterior doors, and skylights shall be tempered glass, *multilayered glazed panels*, glass block or have a fire protection rating of not less than 20 minutes.
- (h) *Underfloor enclosure*. *Buildings or structures* shall have underfloor areas enclosed to the ground with exterior walls in accordance with Sec. 106-51(e).

Exception: Complete enclosure shall not be required where the underside of exposed floors and exposed structural columns, beams and supporting walls are protected as required for exterior one hour *fire-resistance-rated construction* or *mass timber construction* or *fire-retardant-treated wood*.

- (i) *Decks, appendages, and projections*. Decks and other unenclosed *accessory structures* attached to *buildings* shall be constructed of materials per the following:

(1) Deck surface. *Noncombustible* material, *approved* wood thermoplastic composite lumber with an ASTM E84 *flame spread index* no greater than 200, *ignition-resistant building materials* or any *approved Class A roof assembly*.

(2) Deck framing. Deck framing shall be constructed of one of the following:

- (i) *One-hour fire-resistance-rated construction*.
- (ii) *Mass timber construction*.
- (iii) *Approved noncombustible materials*.
- (iv) *Fire-retardant-treated wood*.
- (v) *Ignition-resistant building materials*.
- (vi) Wood with a minimum nominal thickness of at least two inches for joists and four inches for beams and columns or posts. Multi-ply beams and posts with an equivalent nominal four-inch thickness are allowed.

(j) *Vehicle access door perimeter gap*. Exterior vehicle access door shall resist the intrusion of embers from entering by preventing gaps between doors and door openings, at the head, sill, and jam of doors from exceeding 1/8 inch.

Gaps between doors and door openings shall be controlled by one of the following methods.

- (1) Deck surface. Weather-stripping products made of materials that: (a) have been tested for tensile strength in accordance with ASTM D638 (Standard Test Method for Tensile Properties of Plastics) after exposure to ASTM G155 (Standard Practice for Operating Xenon Arc Light Apparatus for Exposure of Non-Metallic Materials) for a period of 2,000 hours, when the maximum allowable difference in tensile strength values between exposed and non-exposed samples does not exceed 10 percent; and (b) exhibit a V-2 or better flammability rating when tested to UL 94 (Standards for Tests for Flammability of Plastic Materials for Parts in Devices and Appliances).
- (2) Door overlaps onto jams and headers.
- (3) Garage door jams and headers covered with metal flashing.

Sec. 106-52. Class 2 intermix structure hardening.

- (a) *General*. Class 2 *structure* hardening shall be in accordance with Sec. 106-52(b) through (c) as well as the provisions of Class 1 *structure* hardening in Sec. 106-51(b) through (j).
- (b) *Protection of eaves and soffits*. Eaves and soffits shall be protected on the exposed underside by one of the following.
 - (1) *Noncombustible* material.
 - (2) *Ignition-resistant building materials*.
 - (3) Materials *approved* for not less than 1-hour *fire-resistance-rated construction*.
 - (4) 5/8 inch type X drywall.

- (5) 2-inch nominal dimensional lumber.
- (6) 1-inch nominal *fire-retardant-treated wood*.
- (7) 3/4 inch *fire-retardant-treated plywood*.

Fascia, where provided and exposed, shall be protected on the backside by one of the following:

- (1) *Noncombustible* material.
 - (2) *Ignition-resistant building materials*.
 - (3) Materials *approved* for not less than 1-hour *fire-resistance-rated construction*.
 - (4) 5/8 inch type X drywall.
 - (5) 2-inch nominal dimensional lumber.
 - (6) 1-inch nominal *fire-retardant-treated wood*.
 - (7) 3/4 inch *fire-retardant-treated plywood*.
- (c) *Exterior doors.* Exterior door shall be *approved noncombustible* construction, solid core wood not less than 1 3/4 inch thick or have a fire protection rating of not less than 20 minutes. Windows within doors and glazed doors shall be in accordance with Sec. 106-51(g).

Exception: Vehicle access doors.

- (d) *Protection of eaves and soffits.* Eaves and soffits shall be protected on the exposed underside by one of the following.

Secs. 106-53 – 106-59. Reserved.”

“ARTICLE V. SITE AREA REQUIREMENTS

Sec. 106-60. General.

- (a) *Scope.* The provisions of this Article shall apply to parcels subject to this code.

Exceptions:

1. For Site and Area Requirements, only Structure Ignition Zone 1 standards apply to the reconstruction, replacement, alteration, or repair of the exterior walls of an existing *building*.
 2. The reconstruction, replacement, alteration, or repair of the existing exterior *roof covering* of an existing *building*.
- (b) *Reference.* As needed, the Wildfire Interface Specialist shall refer to the *Home Ignition Zone (HIZ)* Guide as developed by the Colorado State Forest Service. Where conflicts occur between provisions of this code and the *HIZ* Guide, the provisions of this code shall govern. The provisions of this code, as applicable, shall take precedence over the provisions in the referenced standard.

Sec. 106-61. Class 1 interface requirements

(a) *Structure ignition zone 1 (0-5 feet) - Immediate Zone.*

- (1) Objective. This zone is designated to reduce or eliminate ember ignition and direct flame contact with the *structure*, decks, stairs, and attachments.
- (2) Materials. Use *noncombustible*, hard surface materials in this zone, such as rock, gravel, sand, concrete, bare earth or stone/concrete pavers.
- (3) Plantings. Remove all plantings including shrubs, slash, combustible mulch and other woody debris.
- (4) Trees.
 - (i) There shall be no planting of new trees in the Immediate Zone. Mature trees of no less than 10-inch diameter at 4.5 feet above ground level may be maintained.
 - (ii) *Tree crowns* extending to within 10 feet of the outermost point of any *structure* shall be pruned to maintain a minimum clearance of 10 feet.
 - (iii) Prune tree branches to a height of 6 to 10 feet from the ground or a third of the total height of the tree, whichever is less.
- (5) Storage of combustibles. There shall be no storage of firewood or other combustible materials in this area or areas under decks.

(b) *Structure ignition zone 2 (5-30 feet) - Intermediate Zone.*

- (1) Objective. This zone is designed to give an approaching fire less fuel, which will help reduce its intensity as it gets nearer to *structures*.
- (2) Dead materials. Within the fuel modification area, hazardous dead plant material must be removed from live vegetation.
- (3) Fuels accumulation. There shall be no large accumulations of surface fuels such as logs, branches, slash and combustible mulch. Mulch shall not exceed a maximum depth of 4 inches. Firewood piles must be kept at least 30 feet away from habitable structures unless stored in a *noncombustible* container or structure comprised of *ignition resistant building materials* or *noncombustible* materials.
- (4) Trees. *Tree crowns* extending to within 10 feet of the outermost point of any *structure* shall be pruned to maintain a minimum clearance of 10 feet. Prune tree branches to a height of 6 to 10 feet from the ground or a third of the total height of the tree, whichever is less.
- (5) Shrubs. Shrub groups within this zone shall be spaced to prevent *structure* ignition. Shrubs shall not be planted under trees within the tree drip line.
- (6) Prohibited plants. *Juniperus* spp. (Juniper species), *Thuja occidentalis* (American arborvitae), *Thuja orientalis* (Oriental Arborvitae) are prohibited.
- (7) Slope adjustment of the intermediate zone. For tree removals on any *slopes* of 20% or greater, the *Wildfire Interface Specialist* must provide a report on the appropriate spacing of trees to meet the intent of this code.

- (c) *Retaining walls.* Retaining walls shall be constructed with either *noncombustible* or *ignition-resistant building materials* when any of the following conditions exist
 - (1) The retaining wall is within 8 feet of a *structure* regulated by this code or up to the property line when the property line is less than 8 feet away from the *structure*.
 - (2) The retaining wall is integral to the support of a *structure* regulated by this code.
 - (3) The retaining wall is integral to the egress from a *structure* regulated by this code to a public way, easement, or private road.
- (d) *Fencing.* Fencing within 8 feet of a *structure* regulated by this code or up to the property line when the property line is less than 8 feet away from the *structure* shall be constructed with *noncombustible* or *ignition-resistant building materials*.
- (e) *Site Signage.*
 - (1) Marking of roads. *Approved* signs or other *approved* notices shall be provided and maintained for access roads and driveways to identify such roads and prohibit the obstruction thereof.
 - (2) Marking of fire protection equipment. Fire protection equipment and fire hydrants shall be clearly identified in a manner *approved* by the *code official* to prevent obstruction.
 - (3) Address markers. *Buildings* shall have a permanently posted address, which shall be placed at each driveway entrance and be visible from both directions of travel along the road. In all cases, the address shall be posted at the beginning of construction and shall be maintained thereafter, and the address shall be visible and legible from the road on which the address is located in a manner *approved* by the *code official*.

Sec. 106-62. Class 2 intermix requirements.

- (a) *Structure ignition zone 1 (0-5 feet) - Immediate Zone.*
 - (1) Objective. This zone is designated to reduce or eliminate ember ignition and direct flame contact with the *structure*, decks, stairs, and attachments.
 - (2) Materials. Use *noncombustible*, hard surface materials in this zone, such as rock, gravel, sand, concrete, bare earth or stone/concrete pavers.
 - (3) Plantings. Remove all plantings including shrubs, slash, combustible mulch and other woody debris.
 - (4) Trees.
 - (i) There shall be no planting of new trees in the Immediate Zone. Mature trees of no less than 10-inch diameter at 4.5 feet above ground level may be maintained.
 - (ii) *Tree crowns* extending to within 10 feet of the outermost point of any *structure* shall be pruned to maintain a minimum clearance of 10 feet.
 - (iii) Prune tree branches to a height of 6 to 10 feet from the ground or a third of the total height of the tree, whichever is less.

- (5) Storage of combustibles. There shall be no storage of firewood or other combustible materials in this area or areas under decks.
- (b) *Structure ignition zone 2 (5-30 feet) - Intermediate Zone.*
- (1) Objective. This zone is designed to give an approaching fire less fuel, which will help reduce its intensity as it gets nearer to *structures*.
- (2) Dead materials. Within the fuel modification area, hazardous dead plant material must be removed from live vegetation.
- (3) Fuels accumulation. There shall be no large accumulations of surface fuels such as logs, branches, slash and combustible mulch. Mulch shall not exceed a maximum depth of 4 inches. Firewood piles must be kept at least 30 feet away from habitable structures unless stored in a *noncombustible* container or structure comprised of *ignition resistant building materials* or *noncombustible* materials.
- (4) Trees. *Tree crowns* extending to within 10 feet of the outermost point of any *structure* shall be pruned to maintain a minimum clearance of 10 feet. Prune tree branches to a height of 6 to 10 feet from the ground or a third of the total height of the tree, whichever is less.
- (5) Tree Spacing. *Tree crowns* within this zone shall be spaced to prevent structure ignition and promote fuel discontinuity to limit fire spread.
- (6) Shrubs. Shrub groups within this zone shall be spaced to prevent *structure* ignition. Shrubs shall be at least 10 feet away from the edge of tree branches.
- (7) Prohibited plants. *Juniperus* spp. (Juniper species), *Thuja occidentalis* (American arborvitae), *Thuja orientalis* (Oriental Arborvitae) are prohibited.
- (8) Slope adjustment of the intermediate zone. For tree removals on any *slopes* of 20% or greater, the *Wildfire Interface Specialist* must provide a report on the appropriate spacing of trees to meet the intent of this code.
- (c) *Retaining walls*. Retaining walls shall be constructed with either *noncombustible* or *ignition-resistant building materials* when any of the following conditions exist
- (1) The retaining wall is within 8 feet of a *structure* regulated by this code or up to the property line when the property line is less than 8 feet away from the *structure*.
- (2) The retaining wall is integral to the support of a *structure* regulated by this code.
- (3) The retaining wall is integral to the egress from a *structure* regulated by this code to a public way, easement, or private road.
- (d) *Fencing*. Fencing within 8 feet of a *structure* regulated by this code or up to the property line when the property line is less than 8 feet away from the *structure* shall be constructed with *noncombustible* or *ignition-resistant building materials*.
- (e) *Site Signage*.

- (1) Marking of roads. *Approved* signs or other *approved* notices shall be provided and maintained for access roads and driveways to identify such roads and prohibit the obstruction thereof.
- (2) Marking of fire protection equipment. Fire protection equipment and fire hydrants shall be clearly identified in a manner *approved* by the *code official* to prevent obstruction.
- (3) Address markers. *Buildings* shall have a permanently posted address, which shall be placed at each driveway entrance and be visible from both directions of travel along the road. In all cases, the address shall be posted at the beginning of construction and shall be maintained thereafter, and the address shall be visible and legible from the road on which the address is located in a manner *approved* by the *code official*.

(f) *Structure ignition zone 3 (30-100 feet) - Expanded Zone.*

- (1) Objective. This zone focuses on mitigation that keeps fire on the ground.
- (2) Trees. Prune tree branches to a height of 6 to 10 feet from the ground or a third of the total height of the tree, whichever is less.
- (3) Tree Spacing. *Tree crowns* within this zone shall be spaced at a minimum of 6 to 10 feet.
- (4) Dead materials. Within the fuel modification area, hazardous dead plant material must be removed from live vegetation.
- (5) Fuels accumulation. There shall be no large accumulations of surface fuels such as logs, branches, slash and combustible mulch. Mulch shall not exceed a maximum depth of 4 inches. Firewood piles must be kept at least 30 feet away from habitable structures unless stored in a *noncombustible* container or structure comprised of *ignition resistant building materials* or *noncombustible* materials.
- (6) Slope adjustment of the expanded zone. For tree removals on any *slopes* of 20% or greater, the *Wildfire Interface Specialist* must provide a report on the appropriate spacing of trees to meet the intent of this code.

Secs. 106-63 – 106-70. Reserved.”

Section 2. Should any provision of this ordinance be declared by a court to be unconstitutional or invalid, such decision shall not affect the validity of the ordinance as a whole, or any part thereof, other than the part so declared to be unconstitutional or invalid.

Section 3. This ordinance shall be effective on April 1st, 2026.

INTRODUCED, READ, AND ORDERED PUBLISHED this 3rd day of March, 2026.

PASSED, ADOPTED AND APPROVED this 24th day of March, 2026.